Davide Beltrame

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Education

Bocconi University — Milan, Italy
MSc in CS for Artificial Intelligence – **Grade:** 110/110 *cum laude* — *Sep 2024 – Oct 2026*

- Main coursework: optimization and inference, information theory, computer vision, deep learning, reinforcement learning, computational neuroscience, cryptography, physics of complex systems, partial differential equations.
- Selected for the advanced course track and admitted to two PhD-level courses on statistical and applied ML.
- *Bocconi Students for Machine Learning* (BSML) Advisor & Researcher; *Bocconi Open Source Society* (BOSS) Co-Founder; Consensus Rivista Editor-in-Chief for AI and new technologies.

LUISS Guido Carli University — Rome, Italy
BSc in Management and Computer Science – **Grade:** 110/110 *cum laude* — *Sep 2021 – Jul 2024*

- Full tuition merit-based scholarship; maintained top-class ranking with 8 "laude" grades (31/30).
- **Thesis:** *ML for Eco-climatic Strategy: Predicting Air Transport Emissions.* Supervisor: Prof. Giuseppe F. Italiano.
- Starting Finance Club Guido Carli President (2023-2024); Junior Enterprise LUISS (JELU) Auditor (2023-2024).

Research Experience

Graduate Researcher — Milan, Italy
Bocconi University — *Jun 2025 – Oct 2026*

- **Machine Learning & Statistical Physics Lab (Sep 2025 – Oct 2026):** Studied generalization, memorization, and privacy risk in diffusion models (Stable Diffusion, FLUX) with Prof. Luca Biggio through controlled experiments and real-image benchmarks; co-developed the experimental codebase.
- **NLP research (Jun 2025 – May 2026):** Co-developed zero-shot length prediction for diffusion language models, cutting FLOPs by 60–87% on GSM8K, HumanEval and LongFormQA with no significant performance loss.

Advisor and Researcher — Milan, Italy
Bocconi Students for Machine Learning (BSML) — *Apr 2025 – May 2026*

- Co-first-authored a study on interpretability in medical imaging, comparing CNN saliency maps with 300+ annotations we collected from 14 medical students and doctors.
- Co-organized events on campus (HPC tutorial, meetups with industry) open to the university community.

Selected Publications

- **Co-author:** Garnier-Brun, J., Biggio, L., **Beltrame, D.**, Mézard, M., Saglietti, L. *Biased Generalization in Diffusion Models*. arXiv preprint **(ICML 2026 Spotlight, top 2.2% of submissions)**, 2026.
- **Co-first author:** Rossi, V., Cirò, G., **Beltrame, D.**, Gandolfi, L., Röttger, P., Hovy, D. *Diffusion Language Models Are Natively Length-Aware*. arXiv preprint, 2026. **Accepted poster, DiffuLM Workshop at NeurIPS 2026.**

Work Experience

Technical Consultant — Remote (Italy)
Silverstream AI Inc. — *Sep 2026 – Oct 2026*

- Evaluated AI agent outputs against defined criteria and documented failure modes in structured reports.

AI Engineering Intern — Milan, Italy
CommerceClarity — *May 2025 – Sep 2025*

- Engineered an end-to-end Python pipeline for automated product image compliance, reducing the manual rejection rate from 5% to 0%, unblocking the listing funnel and eliminating human-in-the-loop review costs.
- Designed company-wide analytics dashboards (Laravel/REST/SQL) to expose operational bottlenecks and real-time ingestion metrics, adopted for weekly management reviews.

Tutor — Rome, Italy
Italian Informatics Olympics (OII) Training at LUISS — *Mar 2022 – May 2023*

- Architected the full technical pipeline for competitive programming contests: designed 10+ original algorithmic problems, implemented robust input generators, and wrote reference solvers in Python and C++.
- Prepared lectures on computational complexity, dynamic programming and graph theory.

Skills & Additional Information

Coding & markup: Python (PyTorch, TensorFlow, Pandas, NumPy), C++, C, SQL, MATLAB, LaTeX, HTML.

Tools: Linux, Git, Docker, SLURM (HPC), Weights & Biases, FastAPI, GCP.

Languages: Italian (native), English (fluent C2 – IELTS 8.5), French (intermediate B2), Spanish (basic).

Volunteering: Foundation Banco Alimentare ETS (2019), APS "Il Centro" (2018-2019)

Awards: (district level) national Mathematics, Physics, Informatics and Art History Olympics.

Certifications: GRE General Test (2023): Quant 169/170, Verbal 163/170 (92nd percentile overall).

Programs: McKinsey *New Horizon: Lead What's Next in AI*, Barcelona, Spain (Feb 2026).